

E7 — citations and priority

Obligation (request §4, §5). Retrieve and verify the cited papers and the Erdős #81 record; compare published solutions with Paper IV; separate logical independence, code provenance, mathematical antecedents and bibliographic novelty; decide whether **Theorem C coincides in statement with a result of [15] for fixed rooted defect** and whether §8.1's attribution matches; record later versions of [15] and [5] as of the audit date (2026-09-28).

Target inputs. EN md 71417794... (lines 46, 81–94, 921, 986–1046, 1573–1743, 1789); ES md 8448d14e... (the corresponding lines 46, 987–1043, 1712, 1788 — same content).

Retrieved sources (all saved in `sources/`, SHA-256 in `SOURCES_SHA256.txt`, retrieval 2026-09-28 11:11–11:17–03:00):

Ref	Retrieved	Version status on audit date
[15] Okechukwu, arXiv:2609.20871	abs page + v1 PDF (24 pp., 432 079 B)	Submission history lists only v1 (15 Sep 2026). No later version.
[5] N0zoM1z0/erdos-81	manuscript/main.pdf, README.md, lean/FORMALIZATION_STATUS.md, lean/Erdos81/ExternalInputs.lean, docs/VERIFICATION_STATUS.md at cbde8a0a...; tree listing	main HEAD is cbde8a0a... (2026-09-08T10:47Z); only other branch codex/revise-manuscript-lean-audit at an earlier commit. No later version.
[8] erdosproblems.com/81	problem page, proof-claims thread, discussion thread	Problem status shown OPEN : 4 proof claims (Morluto et al. 2026-09-08 = [5]; Traverso 2026-09-22 = Paper IV v0.8; Traverso 2026-08-29 partial = Paper III; Cipollini 2026-08-10 partial $n^2/6+o(n^2)$).
[9],[12] lean-pool PRs	GitHub API	#348 merged 2026-08-25, commit 540d8e34...; #420 merged 2026-09-13, commit d1de6d2a... Match manuscript.
[2]–[4]	doi.org	10.5281/zenodo.22064657 resolves (record 22064657); concept DOI 21273143 resolves to the same latest record.
[16]	Whitman research page	"Clique Partitions of Split Graphs ... in preparation", no manuscript link. Matches manuscript.
Prior-art search	arXiv API (5 queries, arxiv_search_2026-09-28.txt), 1 web search	Only relevant arXiv hit: [15]. Coverage limit: arXiv + general web search only; no MathSciNet/zbMATH access; forum partial claims found via [8].

Finding E7-F1 — Theorem C's eventual maximum coincides with [15, Theorem 1.1], which the manuscript never cites (MAJOR, attribution; release-blocking)

Literal source ([15] v1 p. 2, rendered `sources/okechukwu_p2.png`). With $F_n = \lfloor n(n+1)/6 \rfloor$ and $Q_s(n) = F_{\{n+s\}} - C_{\{s+1,2\}}$ (eq. 1.1): "Theorem 1.1. For each integer $s \geq 0$ there are constants $K_s \geq 0$ and N_s with the following properties. Every graph G of order n with $\text{rsd}(G) \leq s$ satisfies $\text{cp}(G) \leq Q_s(n) + K_s$. For $n \geq N_s$, the maximum is exactly $Q_s(n)$. Equality at these orders holds precisely for the graphs $(K_{\{k-s\}} \sqcup K_s) \vee K_{\{n-k\}}$, k a nearest integer to $(2(n+s)+1)/6$."

Comparison with Theorem C (EN I. 81–94; Lean `rooted_defect_eventual`, `rooted_defect_maximum`).

Aspect	[15, Thm 1.1]	Paper IV Thm C	Same?						
Graph class	$\text{rsd}(G) \leq s$: in every induced H , outside every proper clique (empty allowed), an s -simplicial vertex ($d - \omega(N) \leq s$)	$\forall U, \forall \text{ clique } R \subsetneq U$ (empty allowed), $\exists v \in U \setminus R$ with a clique $C \subseteq N_U(v)$,	$N_U(v)$	$\leq$	C	+s	Identical ($\omega(N) \geq$)	N	$-s \Leftrightarrow$ such C exists)
Target	$Q_s(n) = F_{\{n+s\}} - C(s+1,2)$	$Q_s(n) := M(n+s) - C(s+1,2)$ (same symbol)	Identical						
Quantifiers / threshold	$\forall s \exists N_s \forall n \geq N_s$	$\forall s \exists N_s \forall n \geq N_s$	Identical						
Upper bound	$\text{cp}(G) \leq Q_s(n)$ eventually (and $\text{cp} \leq Q_s + K_s$ at all orders)	$c_4(G) \leq Q_s(n)$ eventually	Paper IV stronger in piece order ($c_4 \geq \text{cp}$); [15] adds the all-orders form						

Attainment / maximum	"the maximum is exactly $Q_s(n)$ "	"the eventual maximum is $Q_s(n)$, also when cliques of arbitrary order are allowed"	Identical statement
Witness	$(K_{\{k-s\}} \sqcup K_s) \vee K_{\{n-k\}}$ (Lemma 2.5)	C complete (k-s), D independent (s), H independent (n-k), all (CUD)-H links, no C-D edges (E.3)	Same graph family
Equality classification	yes	explicitly not claimed beyond chordal	[15] stronger
Permitted piece orders	cp: unrestricted	upper bound order ≤ 4 ; lower bound unrestricted	Paper IV stronger on upper side

Conclusion (SOURCE_VERIFIED). The *maximum* half of Theorem C coincides in statement with [15, Theorem 1.1]; the *upper-bound* half is [15, Theorem 1.1]'s eventual upper bound strengthened from cp to c_4 . The class definition and the symbol Q_s are those of [15]. (Whether [15]'s method yields c_4 is not asserted by [15]: its localization, Thm 3.3, needs a cutoff $L = L(\epsilon)$; AUDITOR_INFERENCE that the c_4 form is a genuine formal strengthening.)

Manuscript attribution (EN = ES). [15] is cited only as: Cor 1.2 (chordal case, l. 46); Thm 1.3 and Cor 5.5 (stability, l. 46, 1040); Thm 3.3, Lemma 3.4, Lemma 4.1/(4.4) (mechanisms, l. 1038, 1042, 1711); and §8.1 l. 1028: "*Okechukwu [15] studies rooted simplicial defect. Our Theorem C uses that class ... This is the precise scope of the extension incorporated here. Comparing mechanisms is not a priority claim for the extremal formula.*" **[15, Theorem 1.1] is never cited** (grep: zero occurrences of "Theorem 1.1"/"Teorema 1.1" attached to [15] in EN or ES). The abstract (l. 28) and the statement of Theorem C (l. 81–94, where the class is defined as "We say that G has rooted simplicial defect at most s ...") carry no citation of [15] at all.

Why this is material. Theorem C is one of the three headline theorems. A reader of the abstract, introduction and §8.1 is told that [15] studies the class and proves the *chordal* corollary and stability results, but not that [15]'s main theorem already establishes, for every fixed s, the same eventual maximum $Q_s(n)$ with the same extremal family (plus equality classification and an all-orders bound). The disclaimer "not a priority claim for the extremal formula" does not identify the prior theorem. By contrast the manuscript does explicitly acknowledge that [5] proves Theorem B's eventual maximum (l. 46, 1000). The attribution in §8.1 therefore **does not match** the finding requested by §5 of the mandate.

Minimal repair (proposal only, not applied). State at Theorem C (and in the abstract/§1 and §8.1) that [15, Theorem 1.1] proves, for each fixed s, $cp(G) \leq Q_s(n)$ eventually with maximum $Q_s(n)$ and classification of equality graphs, and that Theorem C's contribution relative to [15] is the order- ≤ 4 upper bound (c_4) and the independent formal proof; attribute the definition of rooted simplicial defect and the notation Q_s to [15]. A corrected target requires a new revision ID and a new audit of E7/E6 (request §4.2).

Other E7 checks

ID	Claim (EN line)	Result	Severity
E7-C1	l. 46: [5] establishes the eventual chordal maximum; [15, Cor 1.2] obtains it as a special case	SOURCE_VERIFIED: [5] Thm 1.1 ($\max = M(n)$) and Thm 9.1 ($cp_4 \leq M(n)$, $n \geq N$); [15] Cor 1.2	OK
E7-C2	l. 46/1040: [15, Thm 1.3] asymptotic stability, sublinear defect; [15, Cor 5.5] ϵ - δ edit stability, fixed s	SOURCE_VERIFIED (p. 3, p. 20). Also [15] §7 explicitly asks for a quantitative version of Cor 5.5; Thm 6.1 gives one for $s = 0$ with linear rate 16 δ and hypothesis on c_4 (weaker hypothesis than on cp). The manuscript does not claim exclusivity.	OK (OBSERVATION: the text could note that [15, §7] poses this question)
E7-C3	l. 921: [5]'s FORMALIZATION_STATUS.md declares verification conditional on ExternalInputs.Inputs (Vizing, Häggkvist-Janssen, packing transfer with rational optimality certificates)	SOURCE_VERIFIED at lean/FORMALIZATION_STATUS.md (path is under lean/, not the repo root)	OK / MINOR (path)
E7-C4	Table 5 section locations of [5] (§§2,3,4–5,6,8,9)	Spot-checked: §3 root construction, §4 regularization, §5 local stability, §6 copy path crediting Traverso [TG26a], §8 first entry, §9 main theorem — SOURCE_VERIFIED	OK
E7-C5	l. 1024: [5] credits Paper II for copy scheme and Paper III for split result	SOURCE_VERIFIED ([5] §1, §6 first paragraph)	OK
E7-C6	[5] authorship "Anonymous"	[5]'s PDF prints "ANONYMOUS"; the erdosproblems proof claim names Morluto, Yang Luo, Yinzhen Huang, Grace Lee. The manuscript's policy (retain printed attribution) is transparent. Note: tool [14] Jacobian is by the same group (code-provenance context, disclosed).	OBSERVATION

E7-C7	Antecedents for $n^2/6 + o(n^2)$	Cipollini's partial result (erdosproblems claim 2026-08-10; cited by [5] as [Cip26]) is not cited by Paper IV. Not a dependency; omitted antecedent.	MINOR
E7-C8	I. 48 and Thm A: "answers Erdős Problem 81"	erdosproblems.com lists #81 as OPEN on 2026-09-28 with four unreviewed proof claims. The manuscript's [8] note disclaims certification of status. Claim is the author's mathematical claim, not a record status.	OBSERVATION
E7-C9	[9],[12] PR numbers/merge commits/dates; DOIs; [16]	SOURCE_VERIFIED (table above)	OK
E7-C10	[1] EOZ93, $(1/4-\epsilon)n^2$	Confirmed by [5], [15] and erdosproblems text (secondary); primary CPC article not retrieved (paywall)	UNVERIFIED (primary) — low weight
E7-C11	[17] Behrend / Mathlib <code>Behrend.roth_lower_bound</code>	to be checked in the shared Mathlib source (read-only) — see E3/E4 note	deferred

Logical independence vs provenance vs novelty (separated)

- **Logical independence (Lean):** static reading found no `Erdos81.*` names in the freezer (the `ConeAudit` dynamic check is E4). Paper IV's formal chain does not import [5] or [15] — AUDITOR_INFERENCE from static source; dynamic confirmation pending.
- **Code provenance:** Paper III `Nibble` and Paper II chordal structure are the author's own series; Mathlib's regularity lemma and Behrend bound are third party; Aristotle (Harmonic) assisted the formalization (disclosed). The Jacobian tool [14] belongs to the [5] group (disclosed as a computational tool only).
- **Mathematical antecedents:** Paper II (copy scheme), Paper III (nibble, split case), [5] (same two-regime architecture: mixed K_3/K_4 functional with gains $2/5$, copy path, local stability; [5] predates Paper IV v0.8 by 14 days), [15] (class, $Q_s(n)$, fixed-s maximum, stability), Cipollini (first-order bound).
- **Bibliographic novelty (as of 2026-09-28):** the eventual chordal maximum $M(n)$ is in [5] and [15]; the fixed-s eventual maximum $Q_s(n)$ is in [15]; what remains specific to Paper IV at statement level is the c_4 (order ≤ 4) form of Theorem C, the explicit linear stability constants 16 and 48 (Thm 6.1, Cor 6.1a), the partition-stability statement with a partition-independent root, the explicit tower threshold, and the unconditional (interface-free) Lean formalization.

Gate result

E7 = FAIL — confirmed material attribution defect E7-F1 (literal source evidence above). Other attribution statements checked are accurate. Limitations: arXiv/web only; no human bibliographic database; [1] primary not retrieved.

Rendered from the Markdown semantic source with PyMuPDF; not compiled from the TeX file (no TeX engine installed).